

Experiment No. 4

1. Aim

Use function types, scope, and closures; apply try-catch; build a palindrome checker.

2. Software / Tools Required

- Any text editor (VS Code / Notepad++ / Sublime Text)
- Web Browser (Google Chrome / Mozilla Firefox / Microsoft Edge)
- HTML, CSS, and JavaScript

3. Code

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Vehicle Number Validator</title>
<link rel="stylesheet" href="style.css">
</head>
<body>

<h2>Vehicle Number Validator</h2>
<input type="text" id="vehicleNumber" placeholder="e.g. MH12AB1234" maxlength="10">
<button onclick="checkVehicleNumber()">Validate</button>
<div id="result"></div>

<script src="script.js"></script>
</body>
</html>
```

script.js

```
function validation(regNumber) {
  try {

    if (!regNumber || regNumber.trim() === "") {
      throw new Error("Vehicle number cannot be empty");
    }

    regNumber = regNumber.trim().toUpperCase();

    if (regNumber.length !== 10) {
      throw new Error("Vehicle number must have exactly 10 characters");
    }

    const stateCode = regNumber.substring(0, 2);
    if (!/^[A-Z]{2}$/.test(stateCode)) {
      throw new Error("First 2 characters must be uppercase alphabets (state code)");
    }

    const districtCode = regNumber.substring(2, 4);
```

```

    if (!/^[0-9]{2}$/.test(districtCode)) {
      throw new Error("Characters 3-4 must be digits (district code)");
    }

    const series = regNumber.substring(4, 6);
    if (!/^[A-Z]{2}$/.test(series)) {
      throw new Error("Characters 5-6 must be alphabets (series code)");
    }

    const vehicleNo = regNumber.substring(6, 10);
    if (!/^[0-9]{4}$/.test(vehicleNo)) {
      throw new Error("Last 4 characters must be digits (vehicle number)");
    }

    return { valid: true, message: "Valid vehicle registration number: " + regNumber };
  } catch (error) {
    return { valid: false, message: error.message };
  }
}

function checkVehicleNumber() {
  const input = document.getElementById("vehicleNumber").value;
  const resultDiv = document.getElementById("result");

  const result = validation(input);

  if (result.valid) {
    resultDiv.textContent = result.message;
    resultDiv.className = "valid";
  } else {
    resultDiv.textContent = "Error: " + result.message;
    resultDiv.className = "invalid";
    alert(result.message);
  }
}

```

style.css

```

* {
  box-sizing: border-box;
}

body {
  margin: 0;
  min-height: 100vh;
  display: flex;
  flex-direction: column;
  justify-content: center;
  align-items: center;
  padding: 32px 20px;
  font-family: "Segoe UI", Tahoma, Geneva, Verdana, sans-serif;
  text-align: center;
  background: linear-gradient(135deg, #eef4ff 0%, #f8fbff 45%, #e8fff4 100%);
  color: #1f2937;
}

body::before {
  content: "";
  position: fixed;
  inset: 0;
  background:
    radial-gradient(circle at top left, rgba(37, 99, 235, 0.14), transparent 32%),
    radial-gradient(circle at bottom right, rgba(22, 163, 74, 0.14), transparent 30%);
}

```

```

    pointer-events: none;
}

h2,
input,
button,
#result {
    position: relative;
    z-index: 1;
}

h2 {
    margin: 0 0 22px;
    padding: 24px 28px 10px;
    width: min(100%, 520px);
    font-size: clamp(1.7rem, 4vw, 2.2rem);
    font-weight: 700;
    letter-spacing: 0.03em;
    color: #12325f;
    background: rgba(255, 255, 255, 0.72);
    border: 1px solid rgba(255, 255, 255, 0.7);
    border-bottom: none;
    border-radius: 24px 24px 0 0;
    backdrop-filter: blur(12px);
    box-shadow: 0 12px 35px rgba(18, 50, 95, 0.12);
}

input {
    width: min(100%, 520px);
    padding: 14px 16px;
    font-size: 1rem;
    text-transform: uppercase;
    border: 1px solid #cfd9e8;
    border-radius: 0;
    outline: none;
    background: rgba(255, 255, 255, 0.92);
    color: #1f2937;
    box-shadow: 0 4px 18px rgba(15, 23, 42, 0.06);
    transition: border-color 0.2s ease, box-shadow 0.2s ease, transform 0.2s ease;
}

input::placeholder {
    color: #94a3b8;
    text-transform: none;
}

input:focus {
    border-color: #2563eb;
    box-shadow: 0 0 4px rgba(37, 99, 235, 0.16);
    transform: translateY(-1px);
}

button {
    width: min(100%, 520px);
    margin-top: 14px;
    padding: 14px 18px;
    font-size: 1rem;
    font-weight: 600;
    background: #2563eb;
    color: #ffffff;
    border: none;
    border-radius: 10px;
    cursor: pointer;
    box-shadow: none;
}

```

```

    transition: background-color 0.2s ease;
}

button:hover {
    background: #1d4ed8;
}

button:active {
    background: #1e40af;
}

#result {
    width: min(100%, 520px);
    margin-top: 22px;
    padding: 16px 18px;
    font-weight: 600;
    font-size: 1rem;
    border-radius: 18px;
    background: rgba(255, 255, 255, 0.85);
    border: 1px solid rgba(203, 213, 225, 0.75);
    box-shadow: 0 12px 26px rgba(15, 23, 42, 0.08);
    min-height: 58px;
}

.valid {
    color: #166534;
    background: rgba(220, 252, 231, 0.9) !important;
    border-color: rgba(134, 239, 172, 0.95) !important;
}

.invalid {
    color: #b91c1c;
    background: rgba(254, 226, 226, 0.92) !important;
    border-color: rgba(252, 165, 165, 0.95) !important;
}

@media (max-width: 600px) {
    h2 {
        padding: 20px 20px 10px;
        border-radius: 20px 20px 0 0;
    }

    button {
        border-radius: 10px;
    }

    #result {
        border-radius: 16px;
    }
}

```

4. Output

Vehicle Registration Number Validator

Enter registration number:

AB34CD9873

Validate

✔ Valid Vehicle Registration Number

Mahek Malkan
24070521040

Vehicle Registration Number Validator

Enter registration number:

AB34D987

Validate

✘ Invalid Vehicle Registration Number -
Registration number must be exactly 10
characters long.

Mahek Malkan
24070521040

5. Case Study

caseStudy.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Palindrome Checker</title>

  <link rel="stylesheet" href="caseStudy.css">
  <script src="caseStudy.js"></script>
</head>

<body>
```

```
<div class="container">
  <h2>Palindrome Checker</h2>

  <label for="word">Enter a Word:</label>
  <input type="text" id="word" placeholder="e.g. madam">

  <input type="button" value="Check Palindrome" onclick="checkPalindrome()">

  <div id="result"></div>
</div>

</body>
</html>
```

caseStudy.css

```
*{
  margin:0;
  padding:0;
  box-sizing:border-box;
  font-family: Arial, Helvetica, sans-serif;
}

body{
  background:#f3f7fb;
  color:#333;
  padding:20px;
}

.container{
  width:420px;
  max-width:100%;
  margin:50px auto;
  background:#ffffff;
  border:1px solid #d7e4f0;
  border-radius:8px;
  padding:24px;
}

h2{
  text-align:center;
  color:#4c6b8a;
  margin-bottom:20px;
  font-size:24px;
}

label{
  font-size:15px;
  color:#444;
}

input[type="text"]{
  width:100%;
  margin-top:8px;
  margin-bottom:16px;
  padding:10px;
  border:1px solid #ccc;
  border-radius:4px;
  font-size:15px;
}

input[type="text"]:focus{
```

```

        border-color:#888;
        outline:none;
    }

    input[type="button"]{
        width:100%;
        padding:10px;
        background:#8ba8c4;
        color:#fff;
        border:none;
        border-radius:4px;
        font-size:15px;
        cursor:pointer;
    }

    input[type="button"]:hover{
        background:#7092b3;
    }

    #result{
        margin-top:18px;
        padding:12px;
        background:#eef5fb;
        border-left:3px solid #8ba8c4;
        border-radius:4px;
        display:none;
    }

    #result h3{
        color:#222;
        margin-bottom:8px;
        font-size:18px;
    }

    #result p{
        margin:4px 0;
        font-size:14px;
        color:#444;
    }

```

caseStudy.js

```

// Function to reverse a string
function reverseString(str) {
    return str.split("").reverse().join("");
}

// Closure
function message() {
    let msg = "Palindrome Checker";

    function display() {
        return msg;
    }
    return display;
}

// Main Function
function checkPalindrome() {
    try {
        let word = document.getElementById("word").value.trim();
        let resultBox = document.getElementById("result");
    }

```

```

    if (word === "") {
        resultBox.style.display = "none";
        throw "Please enter a word.";
    }

    if (!/^[A-Za-z]+$/.test(word)) {
        resultBox.style.display = "none";
        throw "Only alphabets are allowed.";
    }

    if (word.length < 3) {
        resultBox.style.display = "none";
        throw "Please enter at least 3 characters.";
    }

    if (word.length > 20) {
        resultBox.style.display = "none";
        throw "Character limit exceeded (20 characters).";
    }

    let input = word.toLowerCase();
    let reverse = reverseString(input);

    let result = "";

    if (input === reverse) {
        result = "Palindrome";
    } else {
        result = "Not a Palindrome";
    }

    resultBox.style.display = "block";
    resultBox.innerHTML =
        "<h3>" + message()() + "</h3>" +
        "<p><strong>Word:</strong> " + word + "</p>" +
        "<p><strong>Result:</strong> " + result + "</p>";

} catch (error) {
    alert(error);
}
}

```

5. Case Study Output

Palindrome Checker

Enter a word or sentence and check if it reads the same backward.

Your text

Check Palindrome

Clear

✔ "level" is a Palindrome.

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Palindrome Checker

Enter a word or sentence and check if it reads the same backward.

Your text

Check Palindrome

Clear

✗ "Mahek" is NOT a Palindrome.

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